Analysis and Requirements

1. Analysis

The **Wikipedia Search and Text-to-Speech Application** is designed to provide users with an easy and efficient way to access information from Wikipedia. The system allows users to search for topics of interest and instantly retrieve relevant articles using the Wikipedia API. Along with displaying the search results, the application includes a Text-to-Speech (TTS) feature that converts article content into spoken audio.

The project addresses the need for quick information retrieval and improved accessibility. Users can either read the retrieved content or listen to it, making the application suitable for different learning preferences and users with visual impairments or reading difficulties. The integration of search and speech functionalities enhances the overall user experience and makes information consumption more convenient.

The system is web-based, responsive, and user-friendly. It focuses on delivering accurate content from Wikipedia while ensuring ease of navigation and accessibility.

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2. Functional Requirements

The system should provide the following functionalities:

1. **Search Functionality**
    - Allow users to enter keywords or topics.
    - Retrieve relevant Wikipedia articles based on the search query.
    - Display article titles, descriptions, and summaries.
2. **Article Retrieval**
    - Fetch information directly from Wikipedia using the Wikipedia API.
    - Display search results in a structured format.
3. **Text-to-Speech Conversion**
    - Convert article text into speech.
    - Provide controls for play, pause, stop, and resume.
4. **User Interface**
    - Simple and intuitive search interface.
    - Responsive design for desktop and mobile devices.

5. **Error Handling**
    - Display appropriate messages when no results are found.
    - Handle network or API errors gracefully.

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3. Non-Functional Requirements

1. **Performance**
    - Search results should be displayed within a few seconds.
    - Speech generation should occur without noticeable delay.
2. **Reliability**
    - Ensure accurate retrieval of data from Wikipedia.
    - Maintain system stability during continuous usage.
3. **Usability**
    - Easy-to-use interface for all categories of users.
    - Clear navigation and readable content display.
4. **Accessibility**
    - Support audio playback for visually impaired users.
    - Provide easy access to speech controls.
5. **Scalability**
    - Support multiple search requests without performance degradation.
6. **Security**
    - Ensure safe communication with external APIs.
    - Protect the application from invalid user inputs.

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4. Software Requirements

- **Operating System:** Windows, Linux, or macOS
- **Frontend:** HTML, CSS, JavaScript
- **API:** Wikipedia API
- **Speech Engine:** Web Speech API (Text-to-Speech)

- **Browser:** Google Chrome, Microsoft Edge, Firefox

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5. Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or higher
- Storage: 500 MB free space
- Internet Connection: Required for accessing Wikipedia data

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6. Feasibility Analysis

Technical Feasibility

The project can be implemented using standard web technologies and freely available APIs, making development straightforward and cost-effective.

Economic Feasibility

The project requires minimal investment since it uses open-source technologies and publicly available APIs.

Operational Feasibility

Users can easily interact with the system without prior technical knowledge, ensuring smooth adoption and usage.

Schedule Feasibility

The project can be completed within a short development timeline, as it involves integrating existing APIs and web technologies.

Conclusion: The Wikipedia Search and Text-to-Speech Application is a feasible, user-friendly, and accessible solution that enhances information retrieval by combining Wikipedia search capabilities with audio-based content delivery.

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